

Reviewer Report

Spectral Isomorphism between Renormalization Flow in Non-Autonomous Quadratic Maps and the Riemann Zeros: A Numerical Study (mca-4499257)

General assessment

The manuscript proposes a numerical construction based on a non-autonomous quadratic map, a logarithmic cooling schedule, a finite-dimensional transfer matrix, and a subsequent comparison of the arguments of its complex eigenvalues with the imaginary parts of the non-trivial zeros of the Riemann zeta function. The combination of a non-autonomous logistic-type map and spectral-phase extraction is unusual and may be of exploratory computational interest.

However, the central conclusions are not supported by the mathematical and numerical methodology presented in the manuscript. In particular, no “spectral isomorphism” is defined or established. The operator whose eigenvalues are actually computed is not the unitary or self-adjoint operator invoked in the Hilbert–Pólya motivation, and the spectrum of the time-averaged matrix used in the computations is not, in general, the spectrum of the underlying non-autonomous evolution. Moreover, the Riemann zeros are used repeatedly in the design, calibration, parameter optimization, and evaluation of the model. The reported agreements are therefore predominantly in-sample fitting results, not independent spectral predictions.

Several additional claims, including the interpretation of the conjugate spectrum, the vanishing fitted intercept, the renormalization-group terminology, and the comparison with ion-trap experiments, are either mathematically unsupported or substantially overinterpreted. There are also internal inconsistencies between the numerical values reported in Table 1 and those reported in Figures 4–5.

These problems affect the fundamental construction not only its exposition. A satisfactory revision would require a new mathematical formulation, a redesigned validation protocol, and a complete repetition of the numerical study. I therefore cannot recommend publication of the manuscript in its present form.

Major concerns

1. The claimed spectral isomorphism is neither defined nor proved.

The title and several passages of the manuscript use the term “spectral isomorphism.” Nevertheless, the manuscript does not define the two mathematical objects that are allegedly isomorphic, the map between them, or the structures preserved by such a map. It is unclear whether the intended concept is an algebraic, topological, unitary, or spectral equivalence. A fitted numerical correspondence between two finite ordered sequences is not an isomorphism.

This issue is already visible in Section 2.1, where the author begins by postulating “a unitary evolution operator U defined on a Hilbert space,” but no Hilbert space, domain, operator formula, or unitarity argument is subsequently provided (Line 96). The numerical object actually constructed in Sections 3.1.1 and 5.2–5.4 is a real, finite-dimensional Markov or Perron–Frobenius-type matrix. Such a matrix is generally non-normal, non-self-adjoint, and dissipative. Its complex eigenvalues lie in or near the unit disk, and their arguments cannot automatically be interpreted as eigenvalues of a Hermitian Hilbert–Pólya operator.

The manuscript does not establish any mathematical relation between the initially postulated unitary operator U and the empirical matrix P used in the computations. Consequently, the Hilbert–Pólya interpretation is not justified by the presented model.

2. The time-averaged matrix does not represent the spectrum of the non-autonomous evolution.

In Module B of Section 3.1.1, the manuscript replaces the non-autonomous sequence of transfer matrices P_t by

$$\bar{P}_T = \frac{1}{T} \sum_{t=1}^T P_t$$

and describes this matrix as a static object that “completely freezes” the dynamical resonance information. A related empirical construction is given in Equation (10) of Section 5.3.

For a non-autonomous discrete system, however, the evolution from time 0 to T is governed by the ordered product

$$P_{T-1}P_{T-2}\cdots P_0,$$

not by the arithmetic mean of the individual matrices. In general, there is no relation between

$$\sigma(P_{T-1}\cdots P_0) \quad \text{and} \quad \sigma\left(\frac{1}{T} \sum_{t=0}^{T-1} P_t\right).$$

The averaging procedure removes the temporal ordering and all non-commutative effects. Two sequences containing the same matrices in different orders have the same arithmetic mean but may generate entirely different propagators.

No adiabatic theorem, ergodic argument, cocycle construction, or convergence theorem is provided that would justify interpreting the eigenvalues of $\bar{P}_T$ as the spectrum of the original non-autonomous dynamics. This is a fundamental issue because the entire spectral comparison is performed on this averaged object.

Furthermore, Equation (10),

$$P_{ij} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{x_t \in B_i, x_{t+1} \in B_j\}},$$

defines joint transition frequencies, not conditional transition probabilities. Indeed,

$$\sum_j P_{ij} = \lim_{T \rightarrow \infty} \frac{1}{T} \#\{t : x_t \in B_i\},$$

which is the occupation frequency of B_i , not 1. A row-stochastic empirical transition matrix would require normalization by the number of visits to the source bin. Since the eigenvalues depend on this normalization, the distinction is not merely notational.

3. The logarithmic cooling law is imposed from the target asymptotics, it is not derived from the dynamics.

Section 5.1 presents the relation

$$\Delta\mu \propto \frac{1}{\ln^2 n}$$

as a theoretical consequence of a “topological isomorphism chain.” The argument combines the known asymptotic zero spacing $\Delta E \propto 1/\ln n$ with the claimed relation

$$\Delta\Phi \propto \sqrt{\Delta\mu}$$

and the imposed correspondence $\Delta\Phi \propto \Delta E$. Importantly, the manuscript itself labels the second identification as a “heuristic correspondence.” (Line 636)

No theorem is given showing that the eigenphase spacing of the discretized transfer matrix has a square-root dependence on the parameter perturbation. The local quadratic form of the scalar map near its critical point does not, by itself, imply the stated perturbation law for the complex spectrum of a finite-dimensional, regularized Perron–Frobenius matrix.

Thus, the $1/\ln^2 n$ schedule is selected precisely to encode the known logarithmic density of the Riemann zeros. It may be used as a modeling ansatz, but it cannot simultaneously be presented as an independently derived dynamical explanation of that density. The additional $k_2/\ln^3 n$ term is likewise introduced to reduce low-order curvature and is subsequently optimized against the target spectrum; it is not derived as a controlled higher-order term of a specified perturbation expansion. The use of “renormalization flow,” “running coupling,” “ultraviolet cutoff,” and related terminology in Sections 3.2.3 and 4.1 is also not supported by a defined renormalization transformation, beta function, fixed-point equation, or universality result. A change in an optimally fitted parameter after changing the grid resolution and the fitting interval is not, by itself, evidence of a renormalization-group flow.

4. The numerical validation contains substantial target leakage and does not demonstrate predictive power.

The Riemann zeros enter the construction at several stages.

First, Section 2.1 states that the critical baseline $\mu_c \approx 1.5437$ is empirically anchored by the first Riemann zero. Second, the eigenphase scale is fixed by the first zero through $k = \gamma_1/\theta_1$. Third, Equation (6) defines

$$\text{ErrSum}(\varepsilon) = \sum_{n=1}^6 |k\theta_n(\varepsilon) - \gamma_n|,$$

and the discretization parameter $\varepsilon = 0.001916$ is selected by minimizing this error over the first six zeros. The subsequent agreement of the same first six zeros in Figure 2 is then described as “lock-in.” This is an optimization result on the training points, not an independent validation.

The same problem becomes more serious in the large- N regime. The manuscript changes the spatial discretization, the numerical construction, the number of iterations, the scaling rule, and the parameters k_1 and k_2 when moving from $N = 100$ to $N = 1000$ and $N = 10000$. Appendix F.3 further permits optimized multi-point scaling and piecewise fitting for large datasets. The reported ability to reproduce the macroscopic counting-function trend is therefore obtained after fitting the model separately to the target range.

A meaningful validation would require the author to fix the complete model, all numerical thresholds, the grid, the cooling law, and all parameters using one training interval, and then predict a disjoint interval of previously unused zeros without any re-optimization, re-anchoring, or piecewise adjustment. No such out-of-sample test is presented.

5. The phase-unwrapping procedure is not mathematically well defined for an unordered matrix spectrum.

Section 5.4 filters the eigenvalues using the ad hoc conditions

$$|\lambda| > 0.4, \quad \text{Im } \lambda > 0,$$

sorts the remaining arguments, and applies phase unwrapping to produce a cumulative phase Φ_n , which is then linearly fitted to the Riemann zeros.

Phase unwrapping has a natural meaning for a continuously sampled phase trajectory or for a single eigenvalue branch followed continuously with respect to a parameter. The eigenvalues of one static matrix, however, form an unordered multiset. Their principal arguments contain no information about how many full rotations should be assigned to one eigenvalue relative to another. The result depends on the chosen ordering, the spectral cutoff, numerical tolerances, branch selection, and possible eigenvalue crossings.

The manuscript provides neither a canonical ordering principle nor a stability analysis with respect to these choices. Since the unwrapping operation creates the monotonically growing quantity that is subsequently fitted to the monotonically growing zero sequence, this ambiguity directly affects the central result.

6. The “vanishing fitted intercept” of the conjugate spectrum is an algebraic consequence of the imposed symmetry.

Sections 3.2.2 and 4.2 attach considerable significance to the observation that including conjugate pairs ($\pm\theta_n$) and performing a free linear fit produces an intercept $b \approx 0$. The manuscript interprets this as spontaneous calibration and as evidence that the negative-energy branch cancels a truncation-induced dispersion.

If the regression dataset contains the symmetric pairs

$$(\theta_i, \gamma_i) \quad \text{and} \quad (-\theta_i, -\gamma_i),$$

then the sample means satisfy

$$\bar{\theta} = 0, \quad \bar{\gamma} = 0.$$

For ordinary least squares, the fitted intercept is therefore identically

$$b = \bar{\gamma} - a\bar{\theta} = 0.$$

This is not a dynamical discovery but an elementary consequence of constructing a centrally symmetric regression dataset. Moreover, the slope of this symmetric free fit is equivalent to the least-squares slope obtained from the positive branch with the intercept constrained to zero.

Accordingly, if Strategies C and D differ only by adding the negative copies of the same positive data, they should produce equivalent slopes and errors. The substantially different values reported in the manuscript indicate that other parts of the dataset, filtering, or fitting protocol have also changed. The experiment therefore does not isolate the effect attributed to conjugate symmetry.

7. The numerical results are internally inconsistent.

Table 1 reports, among other values,

Strategy	k_1	MSE
<i>A</i>	6.761	21.849
<i>C</i>	6.481	8.479
<i>D</i>	4.390	9.467.

In contrast, the caption of Figure 4 gives approximately

<i>A</i>	7.429	20.22
<i>C</i>	8.070	10.08
<i>D</i>	8.783	10.81.

Figure 5 again uses Strategy C with $k_1 = 8.070$, instead of the value 6.481 reported in Table 1.

These differences are too large to be explained by rounding. The manuscript does not state that Table 1 and Figures 4–5 correspond to different random realizations, different parameter sets, different numbers of iterations, or different fitting rules. Consequently, the central ablation study cannot be reproduced or interpreted reliably.

The manuscript also claims extreme sensitivity to floating-point differences of order 10^{-15} over as many as 10^{10} iterations in Appendix F.1. Such sensitivity makes repeated independent runs, random-seed reporting, confidence intervals, high-precision verification, and systematic convergence tests essential. None of these is supplied.

8. The physical comparisons are speculative and do not provide validation of the proposed mechanism.

Figures 3 and 5 compare numerical residuals of the fitted model with error bars from a trapped-ion experiment and emphasize coincident locations near $N \approx 20, 40$, and 76. The manuscript itself acknowledges that the two errors have different sources and that no statistical-significance test or uncertainty budget is provided.

The numerical residual curve displays a substantial systematic drift, whereas the experimental deviations remain concentrated near zero with measurement-dependent uncertainty. Selecting a few indices at which both plots show relatively large excursions does not establish a common mechanism. The comparison is therefore at most an exploratory visual observation and should not be used to motivate conclusions about an intrinsic physical resolution limit, negative-energy cancellation, or experimental hardware design.

The GUE comparison is likewise asymmetric. The established relationship between the Riemann zeros and GUE concerns unfolded local statistics, such as spacing distributions and pair correlations. A single globally scaled GUE spectrum is not expected to reproduce the nonlinear Riemann counting function. Its failure against a model into which the $n/\ln n$ trend has already been encoded does not validate the proposed dynamical mechanism.

Terminology, presentation, and English

The manuscript requires comprehensive rewriting in formal scientific English. The main difficulty is not limited to isolated grammatical errors. The text repeatedly replaces precise mathematical statements with metaphorical or promotional expressions, including “ocean of chaos,” “topological fossil,” “single-sided universe,” “missing negative energy half of the universe,” “structural shock absorber,” “ultimate solution,” “spontaneous calibrator,” “tofu-cutting effect,” and “explodes with a bang.” Such language is inappropriate in a mathematical and computational article and frequently obscures what numerical operation is actually performed.

The manuscript also repeatedly uses non-standard or undefined mathematical terminology, including “Riemann manifold,” “topological stiffness,” “cooling inertia,” “homomorphic mapping,” and “physical topology.” In particular, the non-trivial zeros of the Riemann zeta function, their ordinates, or their counting function are not a “Riemann manifold” in any standard mathematical sense, and no manifold structure is defined in the manuscript. Such expressions should be replaced by the precise mathematical objects actually under consideration.

Several sentences are grammatically incorrect or difficult to parse, and capitalization is inconsistent. Examples include phrases such as “the topological stiffness is instantaneously challenges” and frequent unnecessary capitalization of ordinary technical terms. Even after language editing, however, the primary requirement is to replace metaphorical claims with explicit definitions, equations, algorithms, assumptions, and verifiable statements.

Recommendation

The manuscript contains an unconventional computational idea, but the present results do not establish a spectral isomorphism, a Hilbert–Pólya operator, or an independently predictive relationship between the non-autonomous quadratic map and the Riemann zeros. The main numerical agreements are obtained after target-dependent construction and optimization, while several central interpretations are mathematically invalid or unsupported.

Because resolving these issues would require reconstruction of the mathematical framework and the entire numerical validation, my recommendation is **Reject**.